Unit-V Tutorials

① Mean $= \frac{1}{\mu} = 10$; $\mu = \frac{1}{10}$ / minute $= 6$ / hour

$\lambda = 4$ / hour $\quad \sigma^2 = 0 \qquad \rho = \frac{\lambda}{\mu} = \frac{4}{6} = \frac{2}{3}$

$$L_s = \frac{\lambda^2\sigma^2 + \rho^2}{2(1-\rho)} + \rho = 1.333 \text{ or } 4/3 \text{ cars}$$

$$L_q = \frac{\lambda^2\sigma^2 + \rho^2}{2(1-\rho)} = 0.6667 \text{ or } \frac{2}{3} \text{ cars}$$

$$W_s = \frac{L_s}{\lambda} = \frac{\left(\frac{4}{3}\right)}{4} = \frac{1}{3} \text{ or } 0.3333 \text{ hours}$$

$$W_q = \frac{L_q}{\lambda} = \frac{\frac{2}{3}}{4} = \frac{1}{6} \text{ or } 0.1667 \text{ hours}$$

② $\lambda = \frac{1}{10}$; $\mu = \frac{1}{8}$

$$L_s = \frac{\lambda}{\mu - \lambda} = \frac{1/10}{1/8 - 1/10} = 4 \text{ customers}$$

$$L_q = L_s - \frac{\lambda}{\mu} = 4 - \frac{\frac{1}{10}}{\frac{1}{8}} = 3.2 \text{ customers}$$

$$W_s = \frac{L_s}{\lambda} = \frac{4}{1/10} = 40 \text{ minutes}$$

$$W_q = W_s - \frac{1}{\mu} = 40 - \frac{1}{1/8} = 32 \text{ minutes}$$

③ Number of operators $s = 2$

Call arrival rate 15 /hour or $\lambda = \frac{15}{60} = \frac{1}{4}$ /mins

Mean service time $\mu = 1/2$

Traffic intensity $\rho = \frac{\lambda}{s\mu} = \frac{1/4}{2(1/2)} = \frac{1}{4}$

$$P_0 = \left[\sum_{n=0}^{s-1} \frac{1}{n!}\left(\frac{\lambda}{\mu}\right)^n + \frac{1}{s!}\left(\frac{\lambda}{\mu}\right)^n\left(\frac{1}{1-\rho}\right)\right]^{-1}$$

$$= \left[\sum_{n=0}^{1} \frac{1}{n!}\left(\frac{0.25}{0.5}\right)^n + \frac{1}{2!}\left(\frac{0.25}{0.5}\right)^n\left(\frac{1}{1-0.25}\right)\right]^{-1}$$

$$= \left\{\frac{1}{0!}\left(\frac{0.25}{0.5}\right)^0 + \frac{1}{2!}\left(\frac{0.25}{0.5}\right)^0\left(\frac{1}{1-0.25}\right] + \frac{1}{1!}\left(\frac{0.25}{0.5}\right)^1 + \frac{1}{2!}\left(\frac{0.25}{0.5}\right)^1\left(\frac{1}{1-0.25}\right)\right]^{-1}$$

$$= \frac{2}{3}$$

i) $P(n=2) = \frac{1}{2!}\left(\frac{0.25}{0.5}\right)^2\left(\frac{1}{1-0.25}\right) \times \frac{2}{3} = \frac{1}{9}$

ii) $L_s = L_q + \frac{\lambda}{\mu} = \frac{1}{(s-1)!}\left(\frac{\lambda}{\mu}\right)^s \cdot \left[\frac{\lambda\mu}{(s\mu-\lambda)^2}\right] \cdot P_0 + \frac{\lambda}{\mu}$

$$= \frac{1}{1!}\left(\frac{0.25}{0.5}\right)^2\left[\frac{0.25 \times 0.5}{(1-0.25)^2}\right] \cdot \frac{2}{3} + \frac{0.25}{0.5}$$

$$= \frac{1}{27} + \frac{1}{2} = \frac{29}{54}$$

iii) $L_q = \frac{1}{27}$

iv) $W_s = W_q + \frac{1}{\mu} = \frac{L_q}{\lambda} + \frac{1}{\mu} = \frac{\frac{1}{27}}{1/4} + \frac{1}{1/2} = \frac{58}{27}$

v) $W_q = \frac{L_q}{\lambda} = \frac{1/27}{1/4} = \frac{4}{27}$ or 0.1481

(4) Given $k = 7$; $\lambda = 5$ /hour ; $\mu = \frac{60}{20} = 3$ /hour

$\rho = \frac{\lambda}{\mu} = \frac{5}{3}$

i) $P_0 = \frac{1-\rho}{1-\rho^{k+1}} = \frac{1-5/3}{1-(5/3)^8} = 0.0114$

ii) $L_s = \frac{\rho}{1-\rho} - \frac{(k+1)\rho^{k+1}}{1-\rho^{k+1}} = \frac{5/3}{1-5/3} - \frac{8(5/3)^8}{1-(5/3)^8} = 5.6367$

$L_q = L_s - \frac{\lambda_e}{\mu} = 5.6367 - 1 + P_0 = 4.6367 + 0.0114 = 4.6481$

iii) $\lambda_e = \mu(1-P_0) = 1 - 0.0114\mu = 0.9886\mu = 2.9658$